


期中测验参考答案

一、1. 因为 $f(x) = x^n + nx - 1$

$$f(0) = -1 < 0, \quad f\left(\frac{1}{n}\right) > 0, \quad f'(x) > 0 \Rightarrow f(x) \nearrow$$

$$\Rightarrow 0 < x_n < \frac{1}{n}$$

case 1: $d > 0$, $\sum x_n^n$ 收敛

$$\text{case 2: } d = 1 \Rightarrow x_n = \frac{1 - x_n^n}{n} > \frac{1}{n} - \frac{1}{n^{n+1}} \Rightarrow \text{发散}$$

case 3: $d < 1$, 发散.

2. 直接计算: $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 4\sqrt{u^2 + v^2} \left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2} \right)$$

3. 由 $x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$

$$\frac{1}{\pi} \int_{-\pi}^{\pi} x^4 dx = \frac{1}{2} \left(\frac{2}{3} \pi^2 \right)^2 + 16 \sum_{n=1}^{\infty} \frac{1}{n^4}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90} \quad \left(\text{Parseval} \right)$$

$$4. \quad \prod p_i^{q_i} \leq \prod q_i^{q_i} \Leftrightarrow \sum q_i \ln p_i \leq \sum q_i \ln q_i$$

证明 函数 $f(x_1, \dots, x_n) = \sum q_i \ln x_i$ 在 $x_1 + \dots + x_n = 1$ 上
于 $(q_1, \dots, q_n)$ 处取最大值

$$5. \text{ 通项: } u_n = (-1)^{n-1} \frac{1}{2n-1} \left(1 + \frac{1}{2} + \dots + \frac{1}{n} \right)$$

证明 $|u_n|$ 单调递减即可.

$$6. \quad \begin{cases} f_x = \cos x - \cos(x+y) = 0 \\ f_y = \cos y - \cos(x+y) = 0 \end{cases} \quad \text{于 } D \text{ 上 } \left(\frac{2}{3}\pi, \frac{2}{3}\pi \right) \text{ 为驻点}$$

$$\text{由 } \partial D \text{ 上 } f(x, y) = 0 \text{ 及 } f\left(\frac{2}{3}\pi, \frac{2}{3}\pi\right) = \frac{3\sqrt{3}}{2} > 0$$

$$\Rightarrow f_{\min} = 0 \quad f_{\max} = \frac{3\sqrt{3}}{2}$$

$$7. \quad \sum_{n=2}^{\infty} \frac{k+2}{k! + (k+1)! + (k+2)!} = \sum \left(\frac{1}{(k+1)!} - \frac{1}{(k+2)!} \right)$$

$$= \left(\sum_{k=1}^{\infty} \frac{x^{k+1}}{(k+1)!} - \sum_{k=1}^{\infty} \frac{x^{k+2}}{(k+2)!} \right) \Big|_{x=1} = \frac{1}{2} x^2 \Big|_{x=1}$$

$$= \frac{1}{2}$$

8: $\Sigma: x^2 + y^2 + z^2 = 1$ 外侧单位法向量为

$$\vec{n} = (\cos\alpha, \cos\beta, \cos\gamma)$$

$$\frac{\partial f}{\partial n} = \frac{\partial f}{\partial x} \cos\alpha + \frac{\partial f}{\partial y} \cos\beta + \frac{\partial f}{\partial z} \cos\gamma$$

$$\iint_{\Sigma} \frac{\partial f}{\partial n} dS = \iint_{\Sigma} (x^2 + y^2 + z^2) \frac{\partial f}{\partial n} dS$$

$$= \iint_{\Sigma} \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) (\cos\alpha, \cos\beta, \cos\gamma) dS$$

$$= \iiint_{\Omega} \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \right) dV$$

$$\text{又: } \iint_{\Sigma} (x^2 + y^2 + z^2) \frac{\partial f}{\partial n} dS = \iint_{\Sigma} (x^2 + y^2 + z^2) \left(\frac{\partial f}{\partial x} \cos\alpha + \frac{\partial f}{\partial y} \cos\beta + \frac{\partial f}{\partial z} \cos\gamma \right) dS$$

$$= 2 \iiint_{\Omega} \left(x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} \right) dV$$

$$+ \iiint_{\Omega} (x^2 + y^2 + z^2) \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \right) dV$$

$\partial\Omega = \Sigma$;

$$\Rightarrow I = \frac{1}{2} \iiint_{\Omega} [1 - (x^2 + y^2 + z^2)] \sqrt{x^2 + y^2 + z^2} dV$$

$$= \frac{\pi V}{6}$$

$$\frac{1}{11} = \frac{1}{2} \iint_D f(x,y) dx dy = A$$

$$f(x,y) = x^2 y^2 + xy \int_0^{x^2 y^2} f(x^2 y^2 - t) dt + A$$

$$\iint_D x^2 y^2 dx dy = \int_1^1 x^2 dx \int_{-1}^1 y^2 dy = \frac{2}{9}$$

$$\iint_D A dx dy = 2A \quad \iint_D f(x,y) dx dy = A$$

$$\text{令 } u = x^2 y^2 - t$$

$$\Rightarrow \int_0^{x^2 y^2} f(x^2 y^2 - t) dt = \int_0^{x^2 y^2} f(u) du$$

$$\iint_D \left[xy \int_0^{x^2 y^2} f(x^2 y^2 - t) dt \right] dx dy$$

$$= \iint_D \left| xy \int_0^{x^2 y^2} f(u) du \right| dx dy = 0$$

$$\Rightarrow A = \frac{2}{9} + 2A \quad A = -\frac{2}{9}$$

$$\Rightarrow f(x) = x^2 + x \int_0^{x^2} f(x^2 - t) dt - \frac{2}{9}$$

$$\text{令 } u = x^2 - t \Rightarrow \int_0^{x^2} f(x^2 - t) dt = \int_0^{x^2} f(u) du$$

$$\Rightarrow f(x) = x^2 + x \int_0^{x^2} f(u) du - \frac{2}{9}$$

$$\text{由 } f(1) = 0 \Rightarrow 0 = 1 + \int_0^1 f(u) du - \frac{2}{9}$$

$$\Rightarrow \int_0^1 f(u) du = -\frac{7}{9}$$

$$\Rightarrow \int_0^1 f(x) dx = -\frac{7}{9}$$

$$\text{E} \quad f(1, y) = f(0, y) = f(x, 0) = f(x, 1) = 0$$

$$\int_0^1 x(1-x) \frac{\partial^4 f}{\partial x^2 \partial y^2} dx = \int_0^1 x(1-x) d \frac{\partial^3 f}{\partial x \partial y^2}$$

$$= -2 \int_0^1 \frac{\partial^2 f}{\partial y^2} dx$$

$$\text{同样: } \int_0^1 y(1-y) \frac{\partial^2 f}{\partial y^2} dy = -2 \int_0^1 f(x, y) dy$$

$$\text{于是 } \iint_D x(1-x) y(1-y) \frac{\partial^4 f}{\partial x^2 \partial y^2} dx dy = 4 \iint_D f(x, y) dx dy$$

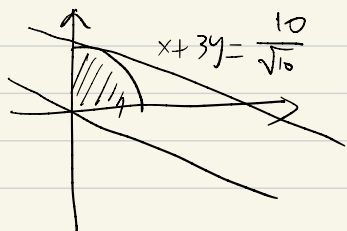
$$\text{由 } \left| \frac{\partial^4 f}{\partial x^2 \partial y^2} \right| \leq 3$$

$$\Rightarrow \left| \iint_D f(x, y) dx dy \right| = \frac{1}{4} \left| \iint_D xy(1-x)(1-y) \frac{\partial^4 f}{\partial x^2 \partial y^2} dx dy \right|$$

$$\leq \frac{3}{4} \left| \iint_D xy(1-x)(1-y) dx dy \right| = \frac{3}{4} \left| \int_0^1 x(1-x) dx \int_0^1 y(1-y) dy \right|$$

$$= \frac{3}{4} \cdot \frac{1}{36} = \frac{1}{48}$$

四：判断 $f(x, y) = \frac{1}{\sqrt{(x+3y+2)^2 + 1}}$ 之极值



$$\text{与 } g(x, y) = x + 3y$$

$$g_{\min}(x, y) = g|_{(0,0)} = 0$$

$$g_{\max}(x, y) = g|_{(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}})} = \frac{10}{\sqrt{10}}$$

$$\Rightarrow \frac{1}{\sqrt{(\frac{10}{\sqrt{10}}+2)^2 + 1}} \leq f(x, y) \leq \frac{1}{\sqrt{2^2 + 1}}$$

进一步由积为保号性

$$\frac{\pi}{\sqrt{(\frac{10}{\sqrt{10}}+2)^2 + 1}} \leq \iint_D f(x, y) dx dy \leq \frac{\pi}{\sqrt{2^2 + 1}} \quad \text{为所证之结果}$$

三 对 $F(x, y, t) = 0 \Rightarrow F(x, y(x), t(x)) = 0$ 求导，

再对 $y = f(x, t(x))$ 求导

解出 $\frac{dy}{dx}$ 即可。

二：交换积分次序即可。